NETWORKING

More Signal, Less Noise

Improving AI Cluster Returns with Enhanced Transceiver Telemetry

Chris Collins

VP of Product, DSP and Optical, Credo

OCP PLATINUM Member

The AI Fabric Visibility Problem

Scale changes the failure model	Bare-metal servers reduce access	Intermittent failures are hard to reproduce
• AI fabrics deploy optical transceivers at unprecedented scale; even rare issues become operationally significant	• Operators often cannot physically inspect or directly query the transceiver on the GPU server side of the link	• Link flaps, LOL/LOS events, and marginal optical paths may disappear before debug begins

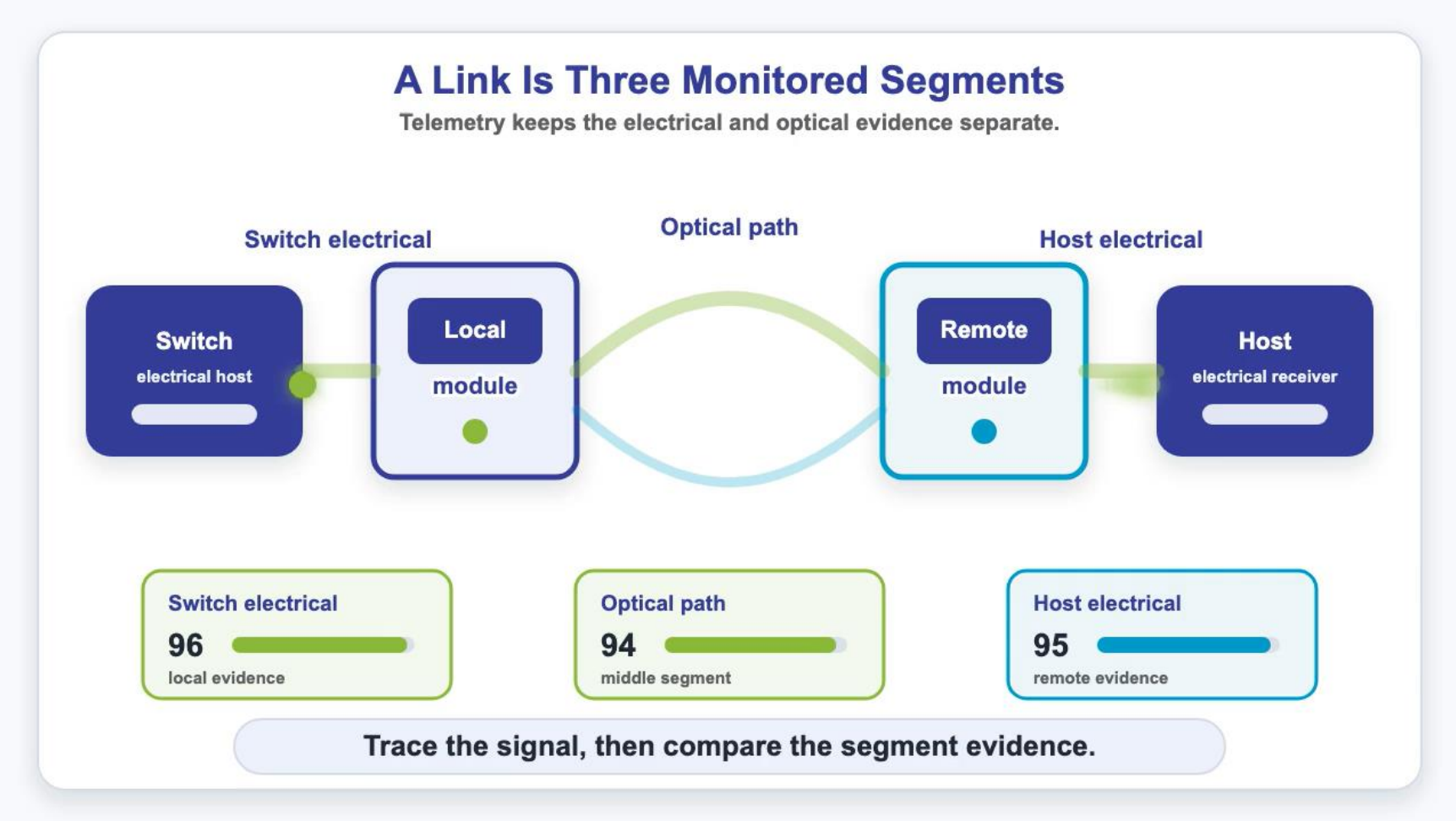
Operators Need to Answer

- Are my links up?
- Who is the link partner?
- Are both ends healthy?
- Which link segment is degraded?
- If a link flapped, why did it flap?



Enhanced telemetry turns optics into intelligent diagnostic endpoints

One Switch View Across The Entire Link



Core Capabilities of Enhanced Telemetry Optics

1. Mission Mode Segmented Link Monitoring

- Continuously tracks host and optical receivers while traffic is live

2. Link Health Reporting

- Normalizes complex telemetry into a simple actionable health indicator

3. Remote Telemetry Harvesting

- Harvests far-end transceiver data from the switch without server access

4. Persistent Event Logging

- Stores important time-stamped events in non-volatile transceiver memory

5. Host Telemetry Access

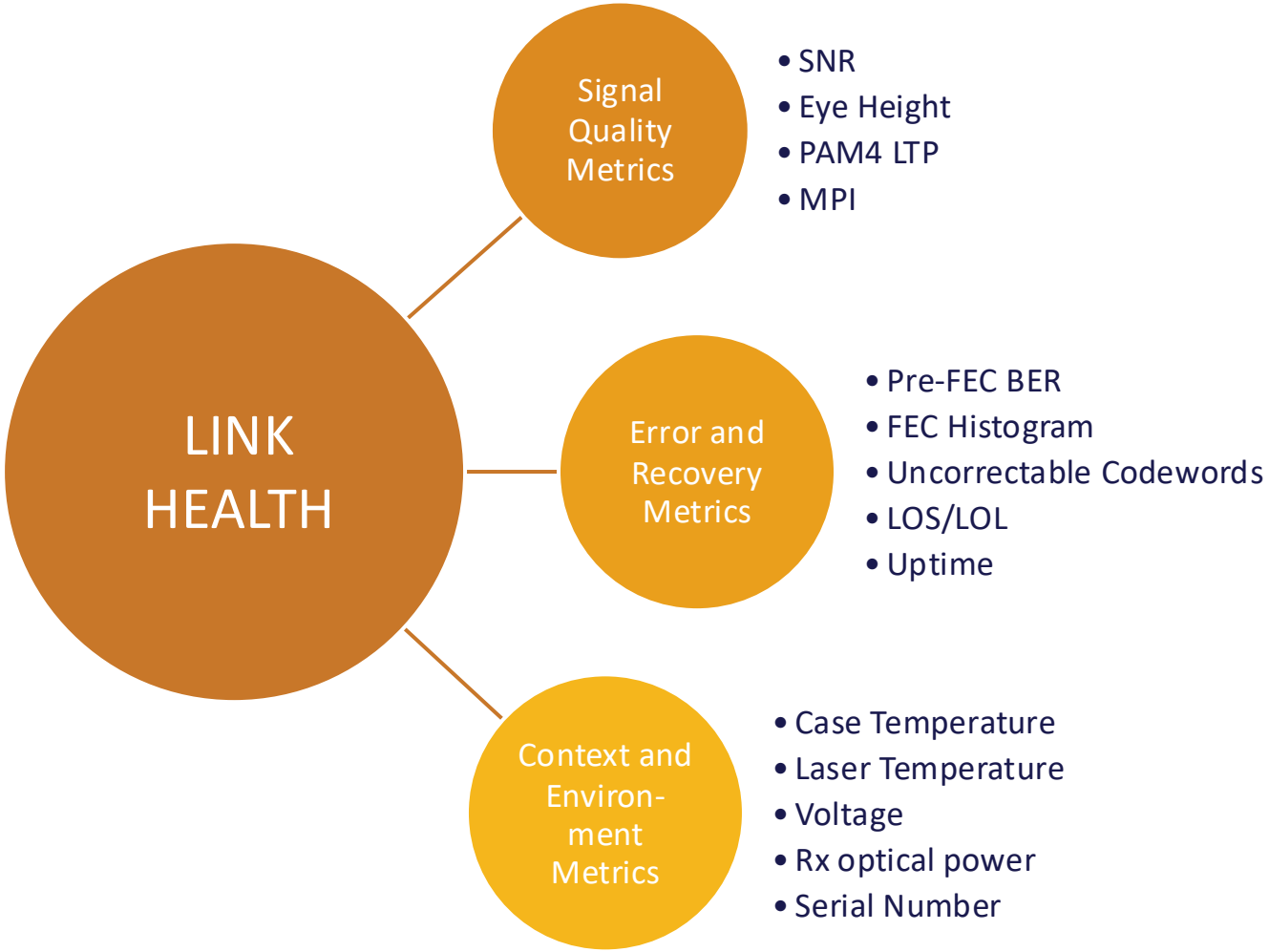
- Provides a common host interface for collection, storage, and analysis

6. Remote Firmware Management

- Supports firmware operations on local and remote modules

Together, these capabilities let operators locate, explain, and remediate optical link issues at AI cluster scale.

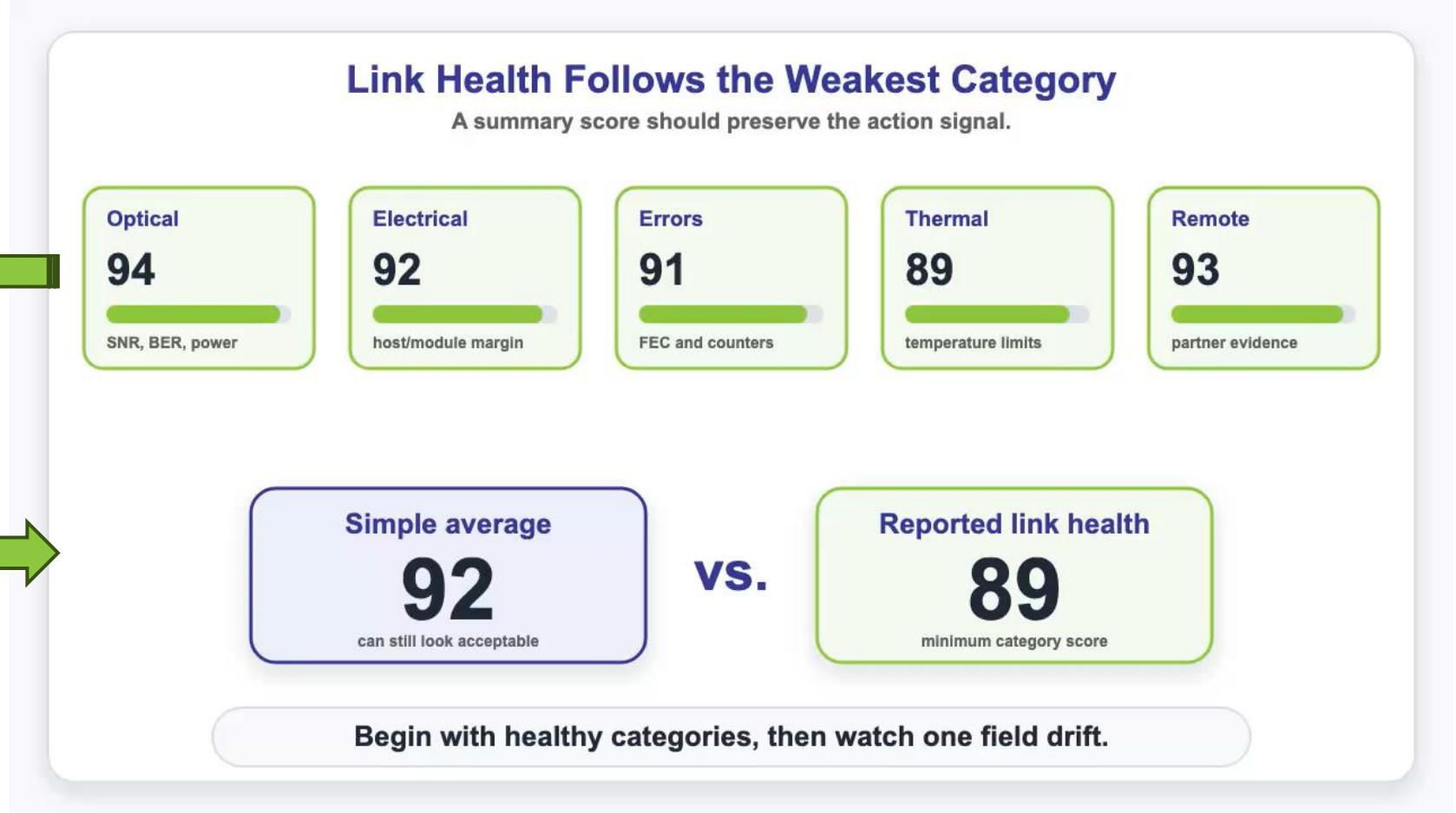
Telemetry Metrics Paint a Picture



Link Health – Follow the Worst Metric

Collect different metrics from each interface

Combine into a single link “Health Score”



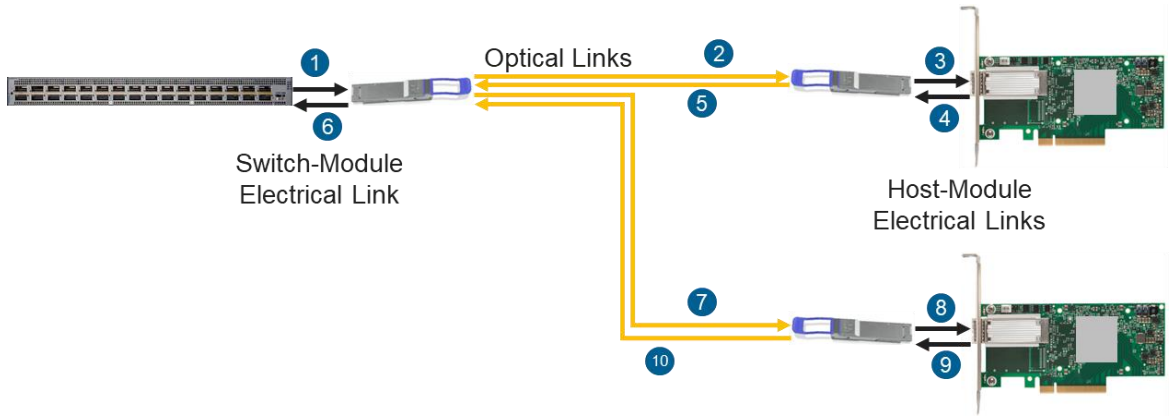
Remote Diagnostics – Without Physical Access

Traditional Optical Transceiver

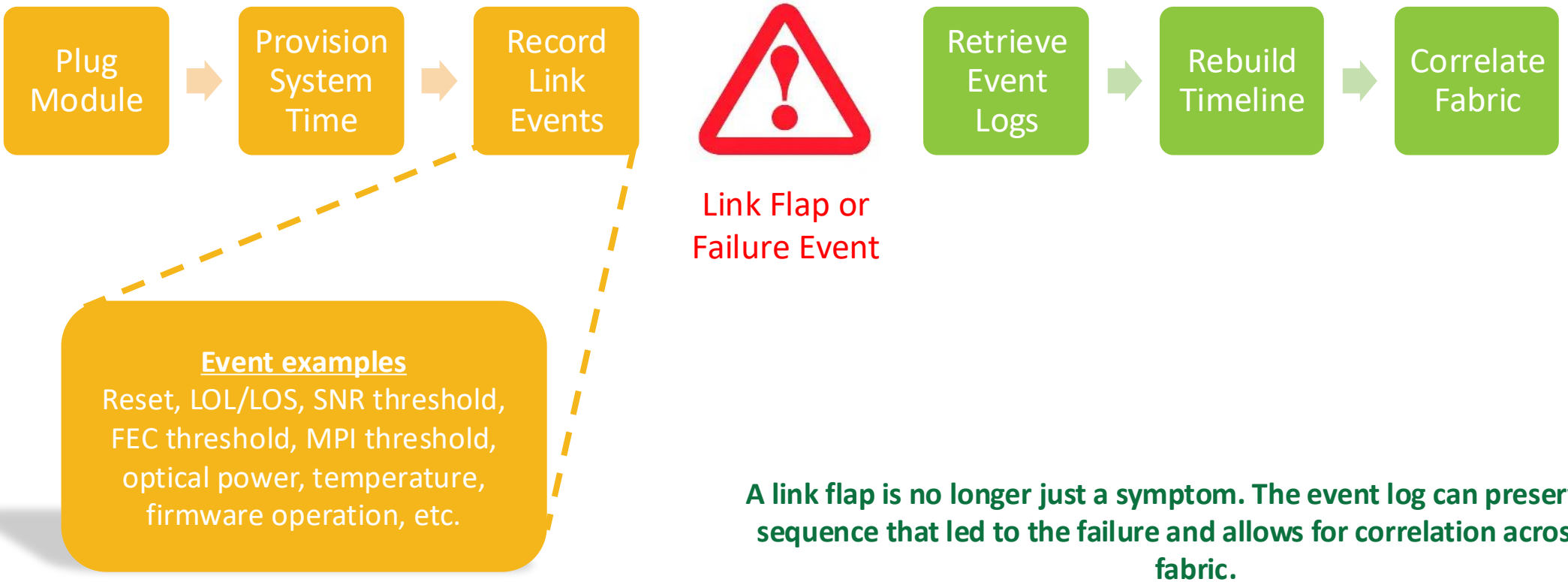
- Query local transceiver only
- Server-side module is a blind spot
- Manual access required for many endpoint checks
- Only partial link visibility

Enhanced Telemetry Transceiver

- Local + remote visibility
- Far-end telemetry and event logs available from the switch
- Works even when GPU servers are bare-metal
- Facilitates full link visibility

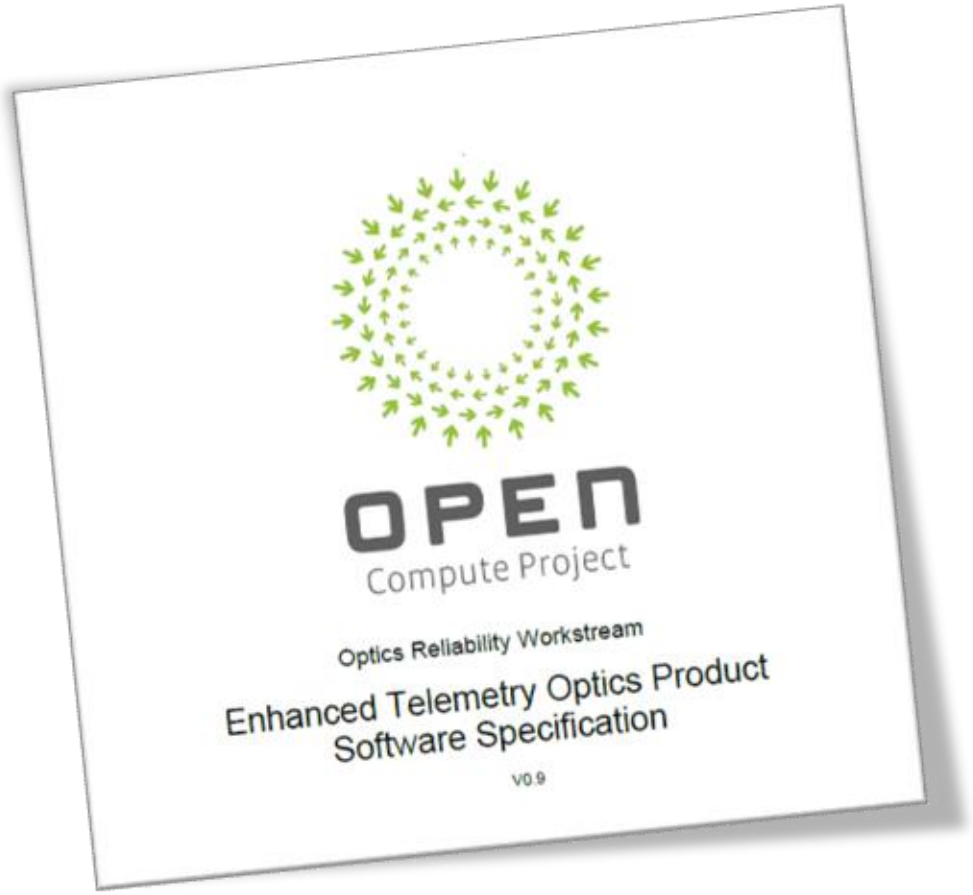


Replay “Time-Machine” with Event Logs



A link flap is no longer just a symptom. The event log can preserve the sequence that led to the failure and allows for correlation across the fabric.

Common Host Interface



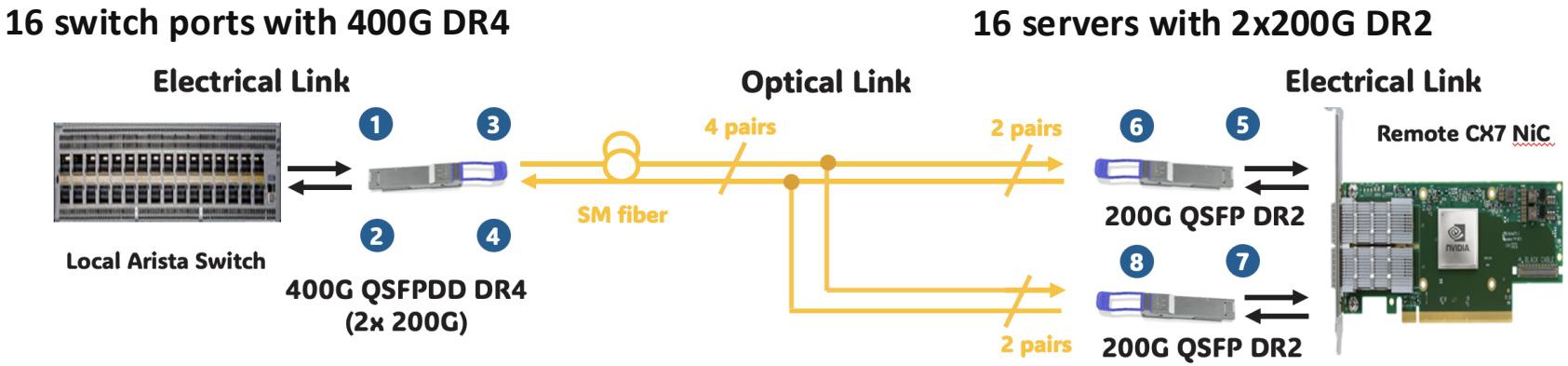
The Enhanced Telemetry Optics Product Software Specification defines a common software interface for compliant transceivers

Built on CMIS CDB, it covers

- Telemetry information
- Link health scoring
- Event log management
- Remote CMIS access
- Remote firmware management

Simplicity by design: a single command returns identity, health score, full telemetry, operating conditions, alarms, and flags

Real World Trial with Credo ZeroFlap Optics



Switch Port Legend

1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31
2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32
33	35	37	39	41	43	45	47	49	51	53	55	57	59	61	63
34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64

No optical attenuation
5dB tx optical attenuation
10dB tx optical attenuation

All 48 transceivers comply with the OCP Enhanced Telemetry Optics Spec

Startup Pre-Check (prior to attenuation)

- One switch port reported a high BER → where's the problem?
- Local and remote telemetry harvesting quickly pin-pointed the electrical side of the server module
- After removal, cleaning, and reseating the identified module, all links are clean

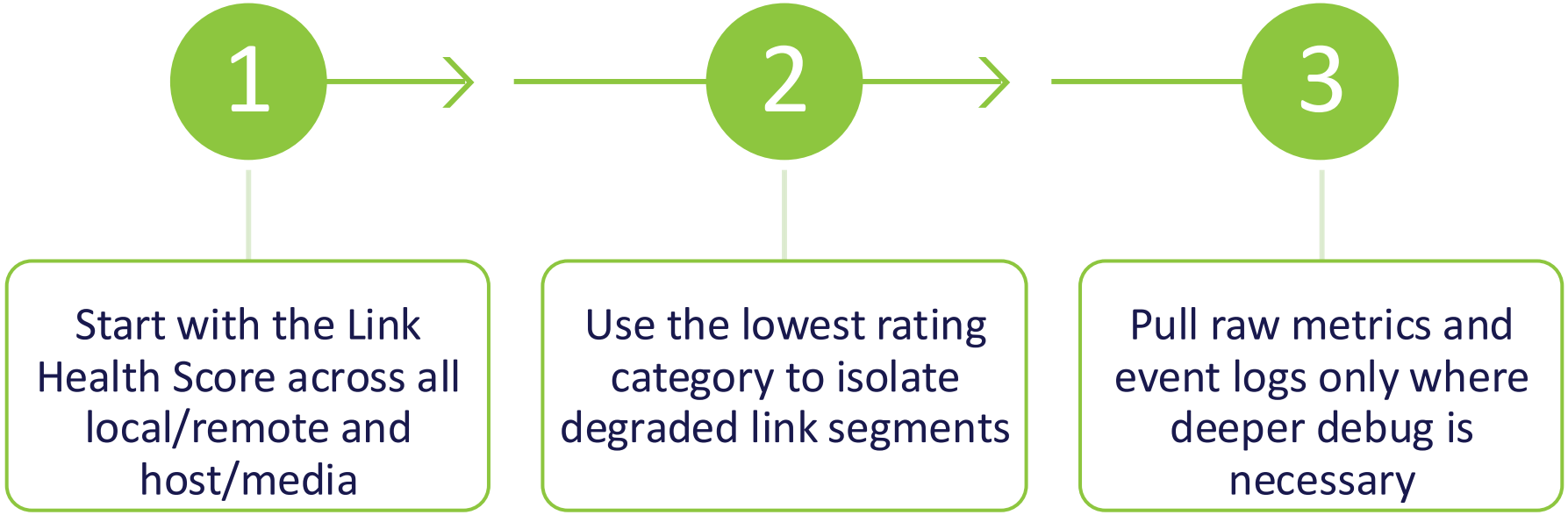
Target	side	Mod Port	Swt Port	Link status	Link Health Score	Lowest Rating Category	Serial number	Link Down Count	MPI	SNR	LTP	Eye height	Temp (C)	Volt (V)
local	host	0	12	TRUE	80	[PRE_FEC_BER, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700031	0	0.50	23.56	56.05	[77, 84, 79]	53.8	3.216
local	media	0	12	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700031	0	0.50	20.88	54.02	[54, 68, 49]	53.8	3.193
local	host	1	12	TRUE	80	[PRE_FEC_BER, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700031	0	0.50	24.13	56.09	[85, 96, 86]	53.8	3.214
local	media	1	12	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700031	0	0.50	20.50	53.91	[53, 61, 48]	53.8	3.191
remote	**host**	**0**	**12**	**TRUE**	**40**	**[EYE_HEIGHT]**	**CQ2AL53600254**	**0**	**0.50**	**17.84**	**52.66**	**[26, 29, 31]**	**40.7**	**3.267**
remote	**media**	**0**	**12**	**TRUE**	**80**	**[MPI, SNR, EYE_HEIGHT, FLAGS]**	**CQ2AL53600254**	**0**	**0.50**	**20.37**	**54.10**	**[62, 62, 52]**	**40.7**	**3.246**
remote	host	1	12	TRUE	80	[PRE_FEC_BER, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53600295	0	0.50	24.81	56.76	[87, 104, 83]	40.31	3.267
remote	media	1	12	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53600295	0	0.50	20.76	54.18	[67, 63, 51]	40.31	3.248

PRE FEC BER	FEC Histogram	Time Stamp	Min Optical Rx Power	Max Optical Rx Power
0	[5294273743862, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]	2025-11-16T07:13:20.332634Z		
		2025-11-16T07:13:34.597162Z	1.60	1.70
0	[5293945137761, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]	2025-11-16T07:13:49.838409Z		
		2025-11-16T07:14:01.198667Z	1.11	1.55
1.50E-06	**[5285133584465, 4294967295, 8657867, 6336, 3, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]**	**2025-11-16T07:13:30.285266Z**		
		2025-11-16T07:13:41.501775Z	**1.27**	**2.08**
0	[5294078325558, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]	2025-11-16T07:13:54.247549Z		
		2025-11-16T07:14:09.322752Z	1.44	1.64

- Local (switch) health looks good
- Suggesting electrical signal issue on server or optical link

- Remote (server) host interface
- Health score suggests an issue.
- Low electrical eye height
- Media side is in good condition

Efficient, Automated, Fabric Wide Fleet Screening



Result: instead of “the port is bad,” teams can say, “remote optical receiver is degraded due to MPI” or “host electrical eye is marginal”.

When automated, this workflow can process an entire fabric in minutes.

Switch Side Media Telemetry – At Startup

- Most ports are in good condition, confirmed by Link Health Score = 80
- One port (18) with a degraded Health Score = 60, due to MPI
  - Typically goes undetected on a traditional optical transceiver during installation
  - In this case, easy to remedy with a fiber cleaning

Target	Side	Module Port	Switch Port	Link status	Link Health Score	Lowest Rating Category	Serial number	Timestamp	Min Optical Rx Power	Max Optical Rx Power
local	media	0	1	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700017	2025-11-16T07:13:37.068294Z	1.26	1.37
local	media	1	1	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700017	2025-11-16T07:14:12.533020Z	1.28	1.66
local	media	0	3	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700022	2025-11-16T07:13:40.590445Z	1.71	1.99
local	media	1	3	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700022	2025-11-16T07:14:19.441604Z	1.50	1.85
local	media	0	5	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700032	2025-11-16T07:13:35.743423Z	1.42	1.71
local	media	1	5	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700032	2025-11-16T07:14:14.577189Z	0.56	0.60
local	media	0	6	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53100318	2025-11-16T07:13:38.915983Z	0.65	0.92
local	media	1	6	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53100318	2025-11-16T07:14:07.473207Z	1.42	1.83
local	media	0	9	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700088	2025-11-16T07:13:35.053050Z	0.50	0.58
local	media	1	9	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700088	2025-11-16T07:14:00.568277Z	1.08	1.75
local	media	0	10	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700100	2025-11-16T07:13:32.365968Z	0.84	1.06
local	media	1	10	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700100	2025-11-16T07:13:59.247849Z	0.44	1.14
local	media	0	11	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53800169	2025-11-16T07:13:36.919870Z	1.07	1.10
local	media	1	11	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53800169	2025-11-16T07:14:03.931331Z	1.21	1.29
local	media	0	12	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700029	2025-11-16T07:13:33.745393Z	0.88	0.94
local	media	1	12	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700029	2025-11-16T07:14:01.920325Z	1.40	1.42
local	media	0	17	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700107	2025-11-16T07:13:34.371109Z	1.95	1.99
local	media	1	17	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700107	2025-11-16T07:13:59.526804Z	0.59	0.80
local	media	0	18	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700016	2025-11-16T07:13:31.755251Z	1.30	1.35
local	media	1	18	TRUE	60	[MPI]	CD4BL53700016	2025-11-16T07:13:58.197781Z	0.99	1.35
local	media	0	19	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700066	2025-11-16T07:13:33.069090Z	0.95	0.99
local	media	1	19	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700066	2025-11-16T07:13:55.573359Z	1.62	2.14
local	media	0	20	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700015	2025-11-16T07:13:30.442102Z	0.24	0.27
local	media	1	20	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700015	2025-11-16T07:13:56.901714Z	0.54	1.31
local	media	0	25	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53100207	2025-11-16T07:13:34.936315Z	-0.16	0.00
local	media	1	25	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53100207	2025-11-16T07:14:00.256350Z	1.07	2.10
local	media	0	26	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700110	2025-11-16T07:13:33.560202Z	1.73	1.94
local	media	1	26	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700110	2025-11-16T07:14:01.944653Z	0.40	0.52
local	media	0	27	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53100279	2025-11-16T07:13:32.219313Z	0.84	1.17
local	media	1	27	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53100279	2025-11-16T07:13:58.915930Z	0.37	0.46
local	media	0	28	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700035	2025-11-16T07:13:36.665765Z	0.52	1.13
local	media	1	28	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CD4BL53700035	2025-11-16T07:14:03.762213Z	1.39	2.32

Server Side Media Telemetry – At Startup

- Different attenuation groups are easy to identify

	Target	Side	Module Port	Switch Port	Link status	Link Health Score	Lowest Rating Category	Serial number	timestamp	Min Optical Rx Power	Max Optical Rx Power
5dB group (except port 9)	remote	media	0	1	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL52700015	2025-11-16T07:13:43.829689Z	-3.08	-2.90
	remote	media	1	1	TRUE	60	[FLAGS]	CQ2AL53600258	2025-11-16T07:14:21.867183Z	-4.33	-3.31
	remote	media	0	3	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53600290	2025-11-16T07:13:50.865637Z	-3.78	-3.23
	remote	media	1	3	TRUE	60	[EYE_HEIGHT, FLAGS]	CQ2AL53600227	2025-11-16T07:14:30.801633Z	-4.15	-3.69
	remote	media	0	5	TRUE	60	[EYE_HEIGHT]	CQ2AL53500118	2025-11-16T07:13:46.937519Z	-3.57	-3.23
	remote	media	1	5	TRUE	60	[FLAGS]	CQ2AL53000131	2025-11-16T07:14:24.945453Z	-4.38	-3.60
	remote	media	0	6	TRUE	60	[FLAGS]	CQ2AL53500141	2025-11-16T07:13:45.585813Z	-4.65	-3.22
	remote	media	1	6	TRUE	40	[FLAGS]	CQ2AL53500063	2025-11-16T07:14:17.180863Z	-6.09	-4.24
	remote	media	0	9	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL52600034	2025-11-16T07:13:41.715474Z	0.79	1.37
	remote	media	1	9	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53500071	2025-11-16T07:14:08.478710Z	0.34	0.94
	remote	media	0	10	TRUE	60	[EYE_HEIGHT, FLAGS]	CQ2AL52700027	2025-11-16T07:13:39.092326Z	-3.95	-3.04
	remote	media	1	10	TRUE	60	[EYE_HEIGHT]	CQ2AL52600094	2025-11-16T07:14:07.162830Z	-3.84	-3.27
	remote	media	0	11	TRUE	40	[FLAGS]	CQ2AL53600210	2025-11-16T07:13:43.848441Z	-5.65	-4.23
	remote	media	1	11	TRUE	60	[FLAGS]	CQ2AL53500087	2025-11-16T07:14:10.843003Z	-4.84	-4.35
	remote	media	0	12	TRUE	60	[FLAGS]	CQ2AL53600239	2025-11-16T07:13:40.431108Z	-4.46	-3.46
	remote	media	1	12	TRUE	60	[EYE_HEIGHT, FLAGS]	CQ2AL53500147	2025-11-16T07:14:05.830301Z	-4.12	-3.72
0dB group	remote	media	0	17	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53600214	2025-11-16T07:13:36.858602Z	1.85	1.97
	remote	media	1	17	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL52600079	2025-11-16T07:14:03.260445Z	1.25	1.95
	remote	media	0	18	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53600298	2025-11-16T07:13:39.451425Z	0.82	0.82
	remote	media	1	18	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL52900336	2025-11-16T07:14:05.833702Z	0.89	1.38
	remote	media	0	19	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53600215	2025-11-16T07:13:40.742203Z	0.89	1.02
	remote	media	1	19	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53500115	2025-11-16T07:14:01.847715Z	0.60	0.90
	remote	media	0	20	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53000081	2025-11-16T07:13:38.146924Z	1.93	2.38
	remote	media	1	20	TRUE	80	[MPI, SNR, EYE_HEIGHT, FLAGS]	CQ2AL53500184	2025-11-16T07:14:03.261072Z	1.66	2.31
10dB group	remote	media	0	25	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL53500122	2025-11-16T07:13:41.538974Z	-9.61	-9.03
	remote	media	1	25	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL52700031	2025-11-16T07:14:05.583229Z	-8.80	-8.41
	remote	media	0	26	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL52900362	2025-11-16T07:13:38.907088Z	-8.27	-7.87
	remote	media	1	26	FALSE	40	[SNR, EYE_HEIGHT]		2025-11-16T07:14:08.254211Z	-40.00	-40.00
	remote	media	0	27	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL53500028	2025-11-16T07:13:38.898688Z	-8.89	-7.89
	remote	media	1	27	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL52700019	2025-11-16T07:14:06.954888Z	-9.14	-8.60
	remote	media	0	28	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL53600250	2025-11-16T07:13:43.630990Z	-8.53	-8.18
	remote	media	1	28	TRUE	40	[EYE_HEIGHT, FLAGS]	CQ2AL53500176	2025-11-16T07:14:10.616954Z	-9.46	-8.71

Link Health Score Tracks Optical Attenuation

0 dB

No attenuation

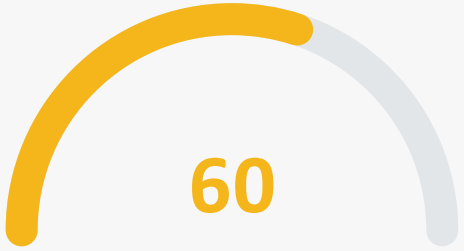


Health Score

Ports 9/17/18/19/20 held good state

5 dB

Moderate impairment

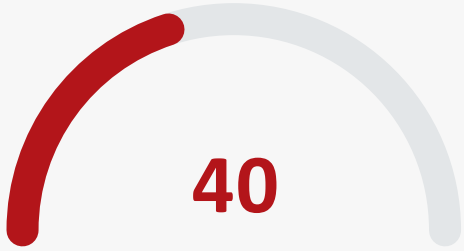


Health Score

Multiple ports warned due to flags or eye height

10 dB

Severe impairment



Health Score

Ports with low Rx power entered critical/alarm state

An easy and effective indicator for massive scale clusters

Call to Action

- How do we bring up AI fabrics faster and keep them running consistently?
  - Use capabilities inside each transceiver to deliver a stream of enhanced telemetry
  - Provide end-to-end visibility of each link
  - Consolidate telemetry from every end-point to deliver a complete fabric health assessment
  - Provide durable root cause evidence for failures
  - Store critical events in memory
  - Promote consistency across vendors to enable broad adoption
- Join the Networking - Optics Reliability Subgroup
  - All contributions and feedback are welcome
- Where to find additional information
  - Project Wiki: https://www.opencompute.org/w/index.php?title=Networking/OpticsReliability
  - Group Home Page: https://ocp-all.groups.io/g/ocp-optics-reliability
  - Mailing list: ocp-optics-reliability@OCP-All.groups.io



Thank You!

